

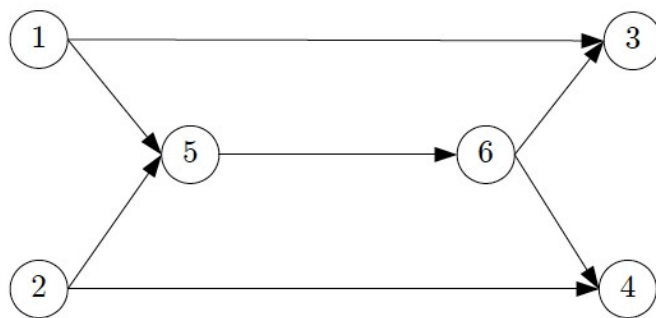
Assignment 1 [100 marks]

Due Date: 5pm Tuesday 28 April 2025

Problem 1 [10 points]

(BLU book, 4.6 Exercises, Q8)

8. [32] Consider the network and OD matrix shown in Figure 4.11. The travel time on *every* link is $10 + x/100$, where x is the flow on that link. Find the link flows and link travel times which satisfy the principle of user equilibrium.



	3	4
1	5,000	0
2	0	10,000

Figure 4.11: Network and OD matrix for Exercise 8.

Solve this problem *analytically*, without using iterative algorithms like the Method of Successive Averages (MSA) and Frank-Wolfe (FW) method.

Problem 2 [30 points]

Consider the network shown in Figure 1. This network has two origin-destination (OD) pairs: OD-pair from Node 1 to Node 5 and OD-pair from Node 2 to Node 5. Travel demands are estimated as 2 flow units for OD-pair 1→5 and 3 flow units for OD-pair 2→5. The link performance function for each link is presented in the figure, where t_{ij} and x_{ij} represent travel time and flow on link (i, j) , respectively. The objective is to perform the User Equilibrium (UE) traffic assignment using the method of successive averages (MSA) and Frank-Wolfe (FW) method.

This question assesses your understanding of the traffic assignment **procedure**, not just the final answer. Present your work in a **clear, handwritten-style format** (scanned handwritten work or equivalent step-by-step manual presentation), clearly showing all intermediate calculations. Follow exactly the procedure illustrated in the lecture exercise for both MSA and FW methods.

Use the following notations:

- Demands: $d^{15} = 2$ (for OD 1→5), $d^{25} = 3$ (for OD 2→5)
- Paths: $\pi_1 = [1,3,5]$, $\pi_2 = [1,3,4,5]$, $\pi_3 = [2,3,5]$, $\pi_4 = [2,3,4,5]$
- Link flow vector $\mathbf{x} = [x_{13}, x_{23}, x_{34}, x_{35}, x_{45}]$
- Link travel times vector $\mathbf{t} = [t_{13}, t_{23}, t_{34}, t_{35}, t_{45}]$

- (a) **[10 marks]** Perform **one** iteration of the MSA algorithm following the initialisation step. Show all steps, including initialisation, convergence checking, and flow updates. Report step size, link flows and costs, relative gap, and average access cost.
- (b) **[15 marks]** Perform **one** iteration of the FW algorithm following the initialisation step. To determine the FW step size at each iteration, use the *bisection* method with interval threshold $\epsilon = 0.2$. Show all steps, including initialisation, convergence checking, and flow updates. Report step size, link flows and costs, relative gap, and average access cost.
- (c) **[5 marks]** Assume a link flow solution has been found, as indicated in Table 1. Assuming that all paths are used, verify if this solution represents the UE traffic assignment results by checking whether it satisfies the principle of UE.

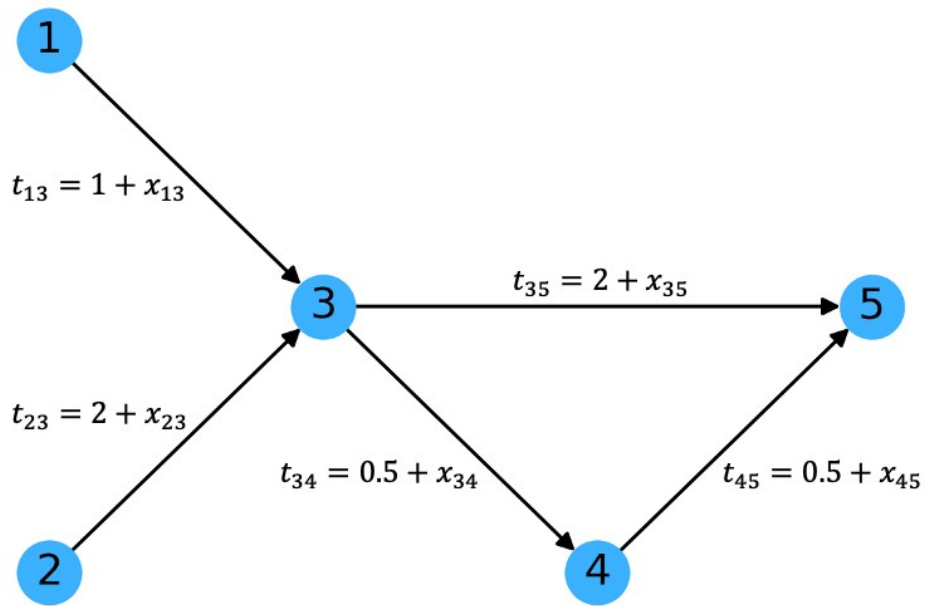


Figure 1

Upstream node i	Downstream node j	Link Flow
1	3	2.0
2	3	3.0
3	4	2.0
3	5	3.0
4	5	2.0

Table 1. Traffic assignment results

Problem 3 [40 points]

Figure 2 below shows a schematic diagram of the **Sioux Falls** network, where the numbers on nodes indicate the node IDs, and the numbers on links indicate the order of the links in the network file ('*SiouxFalls_net.tntp*'). You will evaluate **the impact of building a new highway** in the Sioux Falls network. **The red line connecting nodes 5, 9, 10, 15, 22, and 21 indicates where the new highway is to be built.** Assume that this new highway will **double the capacity** of each of the highlighted links (in both directions) and **reduce the free-flow time by half.**

Your task is to forecast the effects of building this highway by conducting User Equilibrium (UE) traffic assignment using **your Python code** on the SiouxFalls Tntp data. **Your code should follow the structure taught in tutorials in Weeks 4-8**, which cover the complete iterative process to solve a UE traffic assignment problem using the Method of Successive Averages (MSA) and Frank-Wolfe (FW) method. Use the following algorithm **parameters**: *maximum iterations* of **10000**, a *target Relative Gap* (RG) of **0.001** and a *target Average Excess Cost* (AEC) of **0.01**. For the FW method, use a *bisection tolerance* level of **0.0001**.

We will analyse the following three scenarios:

- **Case 1:** Use **MSA** to perform UE traffic assignment for the **original** SiouxFalls network **before** building the new highway
- **Case 2:** Use **FW** to perform UE traffic assignment for the **original** SiouxFalls network **before** building the new highway
- **Case 3:** Use **FW** to perform UE traffic assignment for the **modified** SiouxFalls network **after** building the new highway

(a) **[15 points]** Compare Case 1 vs. Case 2 to evaluate differences between MSA and FW algorithms.

- 1) Report the iteration number, RG, and AEC upon convergence for Case 1 and Case 2, respectively.
- 2) Create the following comparison plots, showing both MSA and FW curves: (1) Step size vs. Iteration, (2) RG vs. iteration, and (3) AEC vs. iteration. Limit the x-axis range to iterations from 0 to 30. Include clear x- and y-axis labels and a legend.
- 3) Discuss the differences in **convergence patterns** between MSA and FW based on the results and plots.

(b) **[25 points]** Compare Case 2 vs. Case 3 to analyse the impact of building the new highway.

- 1) Report the iteration number, RG, and AEC upon convergence for Case 2 and Case 3, respectively.
- 2) Using the traffic assignment results upon convergence, calculate the following network-wide performance measures:

i. **Total System Travel Time:**

$$TSTT = \sum_{(i,j) \in A} t_{ij} x_{ij}$$

ii. **Average Shortest Path Travel Time:**

$$AvgSPTT = \frac{SPTT}{\text{Total Demand}} = \frac{\sum_{(r,s) \in Z^2} \kappa^{rs} d^{rs}}{\sum_{(r,s) \in Z^2} d^{rs}}$$

Compare the values for Case 2 and Case 3 and indicate whether the new highway has increased or decreased these overall network-wide performance measures. Discuss the results.

- 3) Analyse the effects of the new highway on specific links by investigating the following aspects:
 - What links will see increased flow, and what links will see decreased flow?
 - What links will see higher travel times, and what links will see lower travel times?
 - Who are the 'winners' and 'losers' if this highway were to be built?

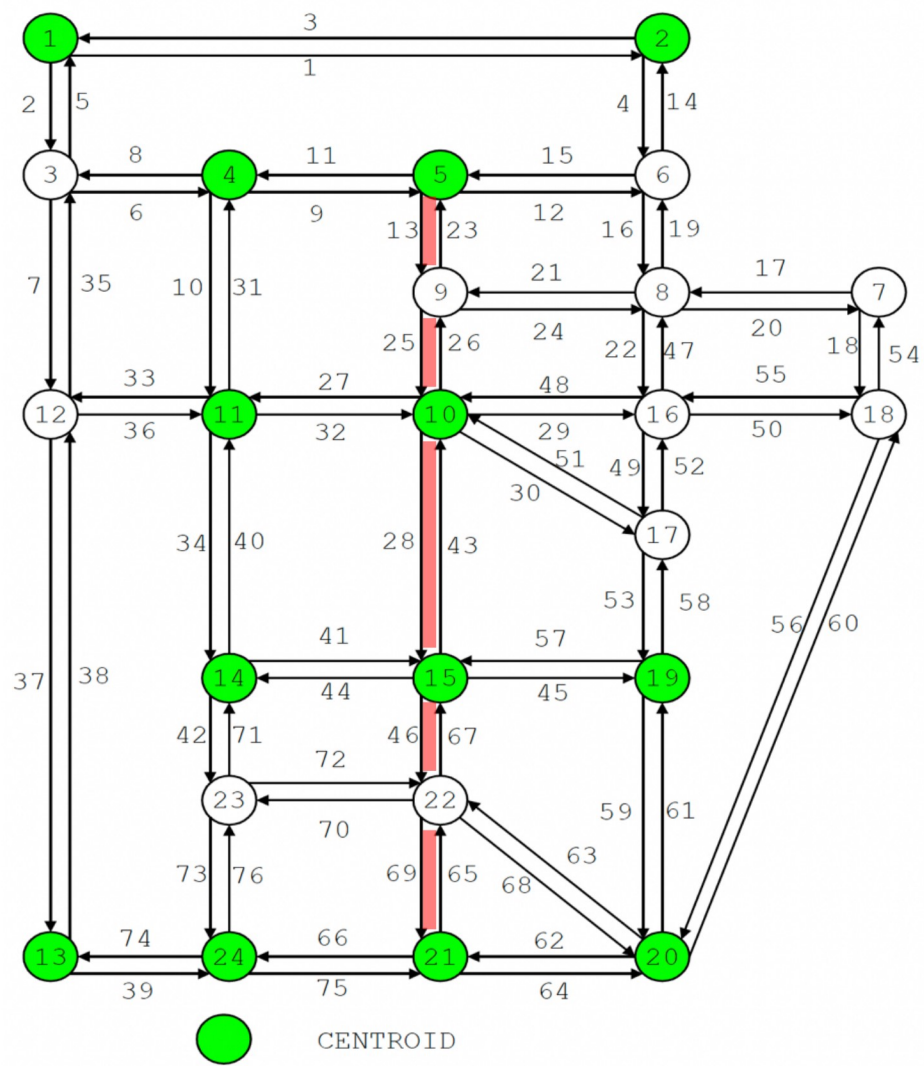


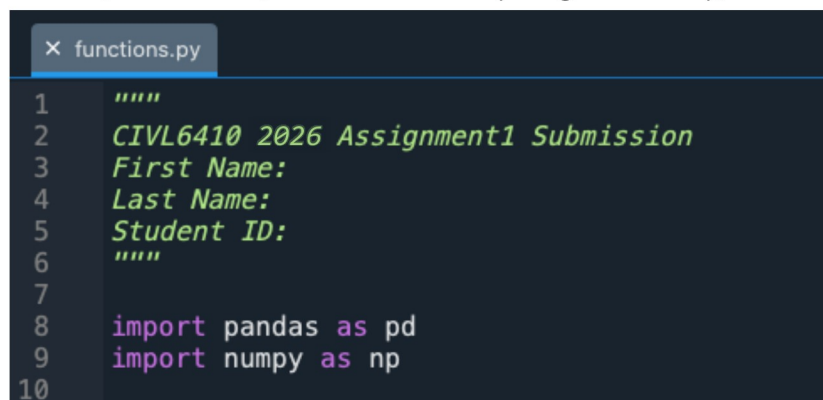
Figure 2. Sioux Falls network

Problem 4 [20 points]

You are required to submit your `functions.py` Python script file, which should contain the following function definitions:

- `shortestPath(orig_node, df_nodes, df_links, forward_star)`
- `updateLinkCost(df_links)`
- `updateOdShortestPathTime(df_nodes, df_links, df_ods, forward_star)`
- `allOrNothingAssignment(df_nodes, df_links, df_ods, forward_star)`
- `calculateConvergenceMeasures(df_links, df_ods)`
- `shiftLinkFlows(step_size, df_links)`
- `getZderivative(step_size, df_links)`
- `getFrankWolfeStepSize(df_links, bisection_tol)`
- `userEquilibriumAssignment(df_nodes, df_links, df_ods, forward_star, max_iter, target_rg, target_aec, method, bisection_tol)`

At the top of your `functions.py` file, include a comment block with your detail including **first name**, **last name**, and **student ID** (8-digit number), as shown below.



```
1  """
2  CIVL6410 2026 Assignment1 Submission
3  First Name:
4  Last Name:
5  Student ID:
6  """
7
8  import pandas as pd
9  import numpy as np
10
```

To grade your code, I will run the `userEquilibriumAssignment()` function in your `functions.py` file using the Sioux Falls TNTF data with specific parameter settings (e.g., `max_iter`, `target_rg`, `target_aec`, `method`, and `bisection_tol`). **Full marks (20 points)** will be awarded only if your function produces the expected results in terms of correct convergence measures and traffic assignment results. **No partial points** will be given if your code fails to execute correctly or produces incomplete or incorrect results.

Submission Instruction

1. Submitting Your Report for Problems 1 - 3

- Your answers to [Problems 1 - 3](#) should be placed in a **single PDF file** (.pdf). Please rename your submission file to follow the format: 'Assignment1_LastName_FirstName_StudentNumber.pdf'.
- Upload the PDF file to the **Blackboard Assignment** submission link:
Assessment > Assignments > Assignment 1 – Report Submission.

2. Submitting Your Code for Problem 4

- Only the '[functions.py](#)' file (a single file) is required for [Problem 4](#). Do not submit any other Python script, data files, or zip files. Do not change the file name; it should be '[functions.py](#)'.
- Upload the Python code ([functions.py](#)) to the **Blackboard Assignment** submission link:
Assessment > Assignments > Assignment 1 – Code Submission.